

DEPARTMENT OF MATHEMATICS
NATIONAL INSTITUTE OF TECHNOLOGY, KURUKSHETRA

B.Tech. III Semester (MAIC-204)
Course Title: Applied Linear Algebra

Mid Term Examination-I

M.M.:15

Time: 6.00 p.m.-6.50 p.m.

Q1. For maps from $\mathbb{R}^2$ to $\mathbb{R}^3$ justify and match the following:

P. $f(x, y, z) = (x, x - y, y)$

Q. $g(x, y) = (x, x, x)$

R. $h(x, y) = (1, x + y, x - y)$

Not a linear transformation.

Rank of linear transformation is 2.

Rank of linear transformation is 1.

(4)

Q2. Justify which of the following are the subspaces of $\mathbb{R}^3$?

a) $\{(x, y, z) \in \mathbb{R}^3 : x - y = 3\}$

b) $\{(x, y, z) \in \mathbb{R}^3 : x + y + z = 1\}$

c) $\{(x, y, z) \in \mathbb{R}^3 : x + y = 0\}$.

(4)

Q3. Consider the following two bases of $\mathbb{R}^2$:

$S = \{u_1, u_2\} = \{(1, 2), (3, 5)\}$ and $S' = \{v_1, v_2\} = \{(1, -1), (1, -2)\}$

(3)

Find the change of basis matrix P from S to basis S'.

Q4. Let V be n-dimensional vector space over the field F, and let W be m-dimensional vector space over the field F. Prove that space $L(V, W)$ is finite dimensional and has dimension mn.

(4)